

Algorithmic thinking: sequence

Revise 360

Worksheet for the 360° experience: Algorithm planning room

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Algorithm planning room. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Explain algorithmic thinking.
- Identify inputs, processes and outputs.
- Write precise ordered instructions.

Starter: before you put the headset on

Put these actions in order: display total, input price, calculate total.

Key terminology

Algorithm

Sequence

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 A clear goal

Turn right

Key facts in your own words:

Challenge: Why state the goal before writing steps?

Algorithmic thinking involves...

A useful algorithm must...

2 Input, process, output

Turn right

Key facts in your own words:

Challenge: Name the inputs for calculating hire cost.

Classify the hire steps.

Categories: Input / Process / Output

Item	Category
Read hours	
Multiply hours by rate	
Display charge	

Multiplying hours by rate is...

3 Order matters

Turn around

Key facts in your own words:

Challenge: What is wrong with printing total first?

Order a total-cost algorithm.

Put these steps in order: Output total / Calculate quantity times unit price / Input unit price / Input quantity

Which must happen before outputting a calculated total?

4 Remove ambiguity

Turn around

Key facts in your own words:

Challenge: Rewrite Add some bonus points precisely.

Which instruction is precise?

Which word often makes an instruction ambiguous?

5 Follow the steps

Turn left

Key facts in your own words:

Challenge: What is printed? Show the intermediate value.

For $x = 5$, then $x = x + 2$, then $x = x * 3$, what is x ?

$x = 3$; $x = x + 4$; $x = x * 2$. Output?

6 Check the result

Turn left

Key facts in your own words:

Challenge: How should the wrong rule be corrected?

A total is wrong. What is the useful next step?

3 items cost 4 each. Why is output 7 wrong?

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Put a simple loan algorithm in order.

Put these steps in order: Display the result / Record the loan if available / Check the item is available / Read borrower ID and item ID

What is the charge for 4 hours at 3 per hour?

Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

Why can instructions that seem obvious to a person confuse a computer?

Review and improve

Use review mode. Explain one correction and why the correct answer fits.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Explain algorithmic thinking. [2]

b) Write an algorithm to input length and width and output area. [3]

c) Explain why the order of instructions matters. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Explain algorithmic thinking.			
Identify inputs, processes and outputs.			
Write precise ordered instructions.			